

BT 505 Current Updated Final Term Spring 2021

Q1. Write down the Features of Glassy Carbon. 5 Marks

Features of Glassy Carbon:

1. High-purity
2. Excellent Stability as high as at 3,000 °C in vacuum
3. Stable at 500 °C in the air
4. Inert against Chemical erosion
5. Impermeability to Gas and Solution
6. Significant Hardness / Strength
7. Favorable electric conducting property
8. Highly Resistant against Inorganic and Organic salts
9. Nonporous
10. Good Bio-Compatibility

Q2. What is the structure of functions of Vitamin B12? What happened to body when its deficiency occurs? 5 Marks

Vitamin B12

1. The structure of B12 is based on a corrin ring
2. The central metal ion is cobalt.
3. There are six coordination site.
4. Four coordination sites are provided by the corrin ring,
5. One coordination site is provided by a dimethyl-benzimidazole group.

Functions and Deficiency

Vitamin B12 performs a key role in the synthesis of DNA, neurolipids, and cellular growth.

It's deficiency causes.

1. Anemia
2. Confusion

3. Depression
4. Problems with balance
5. Poor memory
6. Soreness of the mouth or tongue
7. Tiredness

Q3. Write down the Types of Controller system? 2 Marks

1. Controller system on: Chip All the necessary components are integrated all on-chip.
2. Controller system off Chip: There is a main system to control the functioning of the chip.

Q4. Name of four bioactive molecules use in biosensing layers? 2 Marks

1. Hydrogen peroxidase enzyme
2. Fluorescent Dye
3. Biotin
4. Streptavidin

Q5. Nanoparticle delivery methods into single cells for bioanalysis? 3 Marks

- (a) Endocytosis (non-specific or receptor mediated);
- (b) Pico injection;
- (c) Liposomal delivery;
- (d) Membrane penetration peptide (TAT);
- (e) Gene gun

Q6. What is Gravimetric analysis? What Gravimetric biosensor? 3 Marks

Gravimetric analysis is a class of lab techniques used to determine the

- Mass
 - Concentration of a substance by measuring a change in mass.
1. Gravimetric (mass-sensitive) transduction principles use mechanical acoustic waves.

EFFORT BY AMAAN KHAN

2. Waves are typically generated by means of piezoelectric materials.
3. Signal response changes mainly result from mass changes in the biorecognition layer.
4. Acoustic transduction is label free.
5. Types. Quartz Crystal Microbalance, surface acoustic wave (SAW) and cantilever biosensors.

Q7. How does Urea formaldehyde form? 5 marks

Urea formaldehyde (UF) resins are primarily made up of urea and formaldehyde with formaldehyde acting as the crosslinker.

The UF resins are formed in water at a pH above 7.

Two steps of reaction

1. Formation of the methylol derivatives
2. Condensation:

Q8. How trypsin enzyme activity stabilized? 5 Marks

Trypsin stabilization

Trypsin enzyme activity was stabilized in the presence of Trehalose and was immobilized by using two different techniques. Spray dried immobilization. Freeze dried Immobilization. In each case trehalose act as a stabilizer by forming hydrogen bond with the enzyme and the glass matrix that was used for immobilization.

Q9. What is Biochemical oxygen demand (BOD)? How can we measure the BOD in the industrial area? 5 marks

Biochemical oxygen demand (BOD) is an important index for monitoring organic pollutants in water.

BIOX-1010

1. BIOX-1010 is an on-line analyzer for continuous BOD measurement in industrial and municipal waste-water treatment plants.
2. BOD data are generated with a lag time of only 3 to 15 minutes, with a measurement range of 20 to 1500mg/liter BOD.
3. Inside the bioreactor, microorganisms grow inside small plastic cylinders where they are protected against mechanical abrasion

Q10. Define Fluorophore or Quencher. 3 Marks

Fluorophore is attached at one end of one arm and Quencher is attached at the end of another arm. Quenching is a process which decreases the fluorescence intensity of a substance. When the fluorophore and quencher are in close proximity, the quencher absorbs the energy emitted from an excited fluorophore, thereby suppressing its emission.

Example of Quencher Cl, Acrylamide, NaI.

Q11. Biomarker Characteristics? 3 Marks

Biomarker

1. Biomarker is a biological element.
2. It appears in response to a particular condition.
3. Biomarker is objectively measured and evaluated, as an indicator of
 - I. Normal biological processes,
 - II. Pathogenic processes
 - III. Pharmacological response.

Q12. Stick packs? 3 Marks

Stick Packs

1. Tubular bags, so-called stick packs, are widely used in the packaging industry.
2. It is used in LOC pressure driven devices.

Q13. Formats use in Biosensing? 5 Marks

Types of Test Formats in Biosensor

There are three types of methods that are established in laboratories are used in Biosensor technology.

1. Use of molecular switches: The use of special class of molecules that can exist in more than one stable states with the change in the physical.
2. Immunological methods: Are based on antibody and antigen complex formation on a matrix.

Molecular switches

The molecules may be shifted between the states in response to changes

1. pH
2. Light
3. Temperature
4. Electric current
5. Microenvironment
6. ions and other ligands
7. Combination of stimuli

Immunosorbant Assay

Enzyme linked immunosorbent Assays (ELISA) assay based are one of the formats applied in biosensor measurements.

1. in ELISA antigen (protein) has to be immobilized on a solid support (usually a 96 well plate).
2. After the antigen is immobilized, the detection antibody (primary) is added, forming a complex with the.

Q14. FET? 2 Marks

FET for the detection of EBOLA biosensors

1. Schematic diagram of the rGO-based FET biosensor.
2. rGO sheet was deposited onto the device to bridge the drain and the source electrode.
3. Al₂O₃ was coated on the rGO sheet for surface passivation.

Q15. DNA Damage? 3 Marks

DNA can be damaged by oxidizing agents, alkylating agents and high-energy electromagnetic radiation such as ultraviolet light and X-rays. DNA is damaged due to excessive levels of Reactive Oxygen Species. Normal level of ROS, may be necessary for memory and learning. DNA damaged is related to cancer and other diseases.

Q16. Architecture types on LOCs? 5 Marks

Types

There are different types of architectural design are available for LOC platforms.

1. Lateral flow platform
2. Linear actuated platform
3. Microfluidic large scale integration platform
4. Centrifuged microfluidics platform
5. Electrowetting platform

Linear Actuated Platform

1. The liquid movement is controlled by mechanical devices e.g. by a plunger.
2. Liquid flows in onedimension in a linear fashion without branches or alternative liquid pathways.
3. The reagents and reaction buffers are pre-stored in pouches.

Microfluidic Large scale integration (mLSI)

1. Microfluidic networks to handle large amount of samples.
2. Microfluidic Large scale integration (mLSI) uses microdevices as building blocks.

Centrifugal Microfluidics

1. The microchannels and chambers on substrates are in the format of a disk.
2. The forces involved in the transport of liquid are Centrifugal force, Euler force, Coriolis force and Capillary force.
3. A sequence of liquid operations arranged from radially inward positions to radially outward positions.

Electrowetting Platforms

Electrowetting is the modification of the wetting properties of a surface with an applied electric field. Electrowetting is the reduction of contact angle by an applied voltage.

Q17. Microfluidic chip? 2 Marks

Microfluidic chip is a pattern of molded or engraved microchannels on a chip. To accurately manage fluids inside it is required to understand the identify the inside forces at miniature scale and use a proper setup to run the tests.

Q18. Features Lab on a chip? 3 Marks

Features

1. Safe Platforms: The LOC platforms use very minute amount of samples and reagents, there is no need for any proper training and no specific safety precautions are required.
2. Lower fabrication cost: It allows mass production for cost effective disposable chips.
3. Quality control: The quality of the result is automatically verified.

Q19. How many sites present on Acetylcholine Estrase? Name them. 2 Marks

It possesses three binding sites, i.e.,

1. Active site
2. Aromatic gorge,
3. Peripheral anionic site, where the inhibiting compounds interact.

Q20. How Microfluids maintain equilibrium on a paper? 3 Marks

When a droplet of liquid contacts with paper, the competition of 4 main phenomena.

1. Droplet wets the surface of paper
2. Paper absorbs and wicks the droplet by capillary action
3. Evaporation proceeds
4. Diffusion occurs.

All these four steps lead equilibrium conditions.

Q21. Three ways junction DNA-based and explain its domains? 5 Marks

Three way Junction DNA switch

A series of three-way junction DNA-based molecular switches. It has two domains

1. Recognition domain which interacts with the target of interest and induces a structural change in the TWJ,
2. Transduction domain, which then becomes accessible and binds to a fluorescent molecule.

Q22. Role of Molecular beacon? 2 Marks

Molecular beacons are sensitive fluorescent probes hybridizing selectively to designated DNA and RNA targets. They are used practical tools for quantitative real time monitoring of single stranded nucleic acids. Molecular beacons are ssDNA hybridization probes that form a stem and-loop structure.

Q23. Air pathogen causes? 3 Marks

Diseases caused by Air borne pathogens

1. Respiratory infection,
2. heart disease
3. Lung cancer are the most common causes of premature deaths resulting from air pollution.

Q24. Define biome, biosphere, community. 3 Marks

Biome: A large group of ecosystems defined by soil and climate types.

Biosphere: The global system which includes all life and ecosystems.

Community: A group of interacting populations.

Q25. SELEX Technique. 5 Marks

Sequential Evolution of Ligands by Exponential (SELEX) Enrichment.

Aptamers with affinity for a desired target are selected from a large oligonucleotide library through a process called SELEX.

1. Chromatographic or capillary electrophoresis techniques are used to find the best binding aptamers.
2. Non-binding aptamers are discarded.

SELEX method (Systematic Evolution of Ligands by Exponential Enrichment) was used to select aptamers of Pb (II).

1. Aptamers with high binding affinity and good specificity were selected.
2. A complex is formed with a quencher labeled short complementary sequence and fluorescence labeled aptamer.
3. Detection is based on release of fluorescence-labeled aptamer.

Q26. Microcantilever Array. 5 Marks

The ssDNA1 probes interact with targets in the solution. The microcantilevers bending toward the gold side is identified as bending up (positively), and bending toward the opposite side refers to bending down (negatively). The microcantilever array sensor detection, multiple bacteria can be rapidly and accurately detected at levels as low as 1–9 cfu/ml with excellent specificity.

Q27. DNA probe. 2 Marks

DNA probe is physically labelled with different molecules to improve the performance of the sensor. The binding helps to monitor the DNA hybridization events via an electrochemical method.

Q28. What is difference between Qualitative and Quantitative Biomarker? 2 Marks

Qualitative biomarkers could be involved in a pathogenic process detection within a yes/no analysis.

Quantitative biomarkers are involved in pathogenic process detection with a threshold effect.

Q29. Application of Quantum Dots? 3 Marks

Uses

1. Quantum dots are also being used in place of Organic Dyes in biological research,
2. They can be used like nanoscopic light bulbs to light up and color specific cells that need to be studied under a microscope.
3. Qds being tested as sensors for chemical and biological warfare agents such as anthrax.

Q30. Methods for producing Microdispenser and define the methods to collect the sample for air borne pathogen? 5 Marks

Methods for producing

Microdispenser

There are two method for producing microdispenser.

1. Coldforming method in which no heat is applied and aluminum is used as a substrate.
2. Thermoforming It is manufacturing process. In this method plastic polymers are used and heat is applied. Wet Reagent Storage on LOC Cold Forming
1. In cold forming, an aluminum-based laminate film is simply pressed into a mold by means of a stamp.
2. Al is elongated and maintain the formed shape. Wet Reagent Storage on LOC

Cold Forming

1. In cold forming, an aluminum-based laminate film is simply pressed into a mold by means of a stamp.
2. Al is elongated and maintain the formed shape.

Thermoforming Method

1. In thermoforming method a plastic sheet is heated to a pliable form that is a thermoformed plastic temperature.
2. The warm plastic is treated under pressure (4 to 8 bar) will form the cavity into a negative mold.
3. The mold is cooled such that the plastic becomes rigid again and maintains its shape when removed.

Strategies to collect the sample

Two types of approaches are used to collect air samples that the analytes molecules may be capture along the matrix, that is air.

1. Use of different filters to collect the pathogen
2. Use of gravitational collector

Q31. Juvenile serum lead levels? 2 Marks

The inhibition of brain development, allowable juvenile serum lead levels are just 100 parts-per-billion (ppb).

Q32. Mechanisms of intrinsic antibiotic resistance in *P. aeruginosa*. 3 Marks

Mechanisms of intrinsic antibiotic resistance in *P. aeruginosa*.

1. Restricted outer-membrane permeability
2. Efflux systems that pump antibiotics out of the cells
3. Production of antibiotic inactivating enzymes.

Q33. Advantages of Conductometric biosensors? 5 Marks

Conductometric biosensors have important advantages:

1. They do not need the use of a reference electrode.
2. They are insensitive to light.
3. They can be miniaturized.
4. They can be integrated easily using a cheap thin-film standard technology.

Q34. Explain the Apparatus of Capillary Electrophoresis? 5 Marks

Equipment

1. Capillaries are typically of 25-75 micro meter inner diameter.
2. The applied potential is applied. Electroosmotic flow is from negative to positive electrode.
3. A small volume of sample (10 nL) is injected at the positive end of the capillary and the separated components are detected near the negative end of the capillary.

Q35. Hygroscopic character of cellulose paper? 3 Marks

The paper sensor exploits the hygroscopic character of cellulose paper, that is

1. The tendency to adsorb water from the environment.
2. At a relative humidity (RH) of 70%, paper absorbs water that is up to 10% of its weight.
3. Under normal environment paper contains 10% of water by weight.

Q36. What is Array? 2 Marks

An array is a data structure that contains a group of elements. The collection of Elements have

- Same Name
- Same Data type

Q37. Differentiate between Beta and Gama rays? 2 Marks

Beta particles carry a negative charge. Beta particles are high energy electrons.

Gamma rays are neutral. Gamma rays are waves of electromagnetic energy, or photons.

Q38. What are requirements for microfluidic? 3 Marks

The elements of microfluidic chip are required to accurately manage the fluid inside.

1. The components can be embedded inside the chip or outside the chip.
2. The Quake pneumatic microvalves could be fitted inside.
3. The pressure controllers are fitted outside the microfluidic chip.

Q39. Explain Traditional analytical methods of environmental pollutants.

What is the reason Failure of Traditional methods? 5 Marks

Traditional analytical methods employed for the environmental monitoring of pollutants include various

1. Chromatographic techniques
2. Gas chromatography and
3. High performance liquid chromatography coupled with capillary electrophoresis
4. Mass spectrometry

They require expensive reagents, time consuming sample pre-treatment, and expensive equipment.

Failure of Traditional methods

The traditional methods are not effective for in real time measurements, For example as in the case of accidental release of pesticides or acute poisoning where:

1. Rapid,
2. Miniaturized,
3. Portable equipment is needed such as environmental monitoring biosensors.

Q40. What is Paper Chromatography? 3 Marks

Paper Chromatography

Paper Chromatography is a method of separating mixtures by using a moving solvent on filter paper. A drop of mixture is spotted at one end of the paper and then dried. A solvent system is used for the separation of different compounds on the basis of their mobility.

Q41. What are different methods of assembly of nucleic acid? 2 Marks

Modes of assembly:

1. Native RNA structures
2. Natural scaffolds
3. Synthetically programmed
4. Synthetic scaffolds

Q42. Functions of Ethylene? 5 Marks

The monitoring of the freshness of climacteric fruits is very important. It is monitored by measuring plant hormones. Ethylene is a gas that serves as a hormone in plants.

It plays a critical role in plant growth and development such as

1. Stimulating of fruit ripening
2. Regulating the ripening of fruit
3. Opening of flowers
4. Shedding of leaves

Q44. Wet Etching and Plasma Etching? 3 Marks

Wet Etching:

Hydrofluoric acid is used to bath the treated silicon wafers for wet etching.

Plasma Etching:

Standard conditions for such as pressure under vacuum is created for etching.

Q45. Refractive index? 3 Marks

Refractive index(n) is defined as the ratio of the speed of light in a vacuum to the speed of light in another substance. This property is used to assess composition or purity of a substance.